

Name: _____



Using the fx-CG50 for number and algebra

Task A: Solving equations graphically

1) Use your calculator to solve these pairs of simultaneous equations.

a) $y = x - 3$

$$y = 3x + 6$$

b) $6y - x = 14$

$$x + 3y = 13$$

2) For each pair of simultaneous equations, write down what your calculator says when you try to solve them graphically. Why do you think the calculator gives this output?

a) $35x - 7y = 63$

$$y = 5x + 1$$

b) $3x + 4y = 7$

$$y = \frac{4}{3}x - 7$$

c) $25x - 5y + 5 = 0$

$$y = 5x + 1$$

3) Use your calculator to estimate the solutions to this pair of simultaneous equations to 3 d.p.

$$y = 3x - x^2$$

$$y = 1.7 - 0.8x$$

Task B: Solving equations numerically

1) Solve the following pairs of simultaneous equations using your calculator.

a) $2a + 3b = 1$

$$-a + 4b = 8$$

b) $2b + a = 8$

$$2b + 3 = 7a$$

2) Calculate the solutions to these quadratic equations.

a) $-x^2 - 8x + 20 = 0$

b) $x^2 + 8x - 19 = 0$

c) $x^2 + 6x + 8 = 3$

d) $5x^2 - x + 5 = 3x^2 - 10x - 2$